Using Microsoft Excel to Access Data from Informatica Data Vault

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Abstract

Informatica Data Vault (IDV) is a compressed read-only database that stores inactive data. This article describes the steps to use Microsoft Excel to read data from Data Vault.

Supported Versions

- Informatica Data Vault 6.1 onwards
- Microsoft Excel 2010 onwards

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Overview

Data Vault is a compressed read-only database that the Data Archive product uses to store retired and archived data. With Microsoft Excel, you can access data from the Data Vault. You can perform data analytics on the retrieved data.

Using Microsoft Excel for Querying Data Vault

To access data from the Data Vault, perform the following steps:

1. Install the Data Vault ODBC drivers.
2. Create the ODBC Data Source Name (DSN) for the Data Vault archive folder.
3. Use the ODBC data source in Microsoft Excel to retrieve data from the Data Vault.

Step 1. Install the Data Vault ODBC Drivers

Install the ODBC driver on the machine from where Microsoft Excel can access the Data Vault.

1. Download the latest Data Vault installer.

   If the Data Vault installer is not available, contact the Informatica Shipping Team to get installer links for you to download.
2. Transfer the installer to the Windows client machine, where you can use Microsoft Excel to connect to the Data Vault.

3. Launch the Data Vault installer.
4. On the Introduction page, read the instructions and click **Next**.
5. On the Installation Type page, select **Install a Data Vault component** and click **Next**.
6. On the License File page, enter the path and file name of the Data Vault license key or click **Choose** to select a license key file.
7. On the **Components to Install** page, select **Data Vault ODBC Driver** and click **Next**.
8. On the **Installation Directory** page, enter the absolute path for the directory in which to install the Data Vault ODBC driver or click **Choose** to select or create a directory.

   The target installation directory must be empty. The path must conform to the length and character restrictions of the operating system.
9. Click **Next**.
10. On the **Pre-installation Summary** page, review the installation information, and click **Install** to continue.

    The installer copies the Data Vault ODBC driver files to the installation directory. After the installation process completes, the **Installation Complete** page indicates whether the installation completed successfully.
11. Click **Done**.

The installer creates a log file named *FAS-installer.log* in the root of the directory that contains the installer files. You can view the installer log file to get more information about the tasks that are performed by the installer and to view configuration properties for the installed component.

For more detailed instructions about performing installation of IDV ODBC driver, please refer Informatica Data Archive Installation and Upgrade Guide available in Informatica Knowledge Base site.

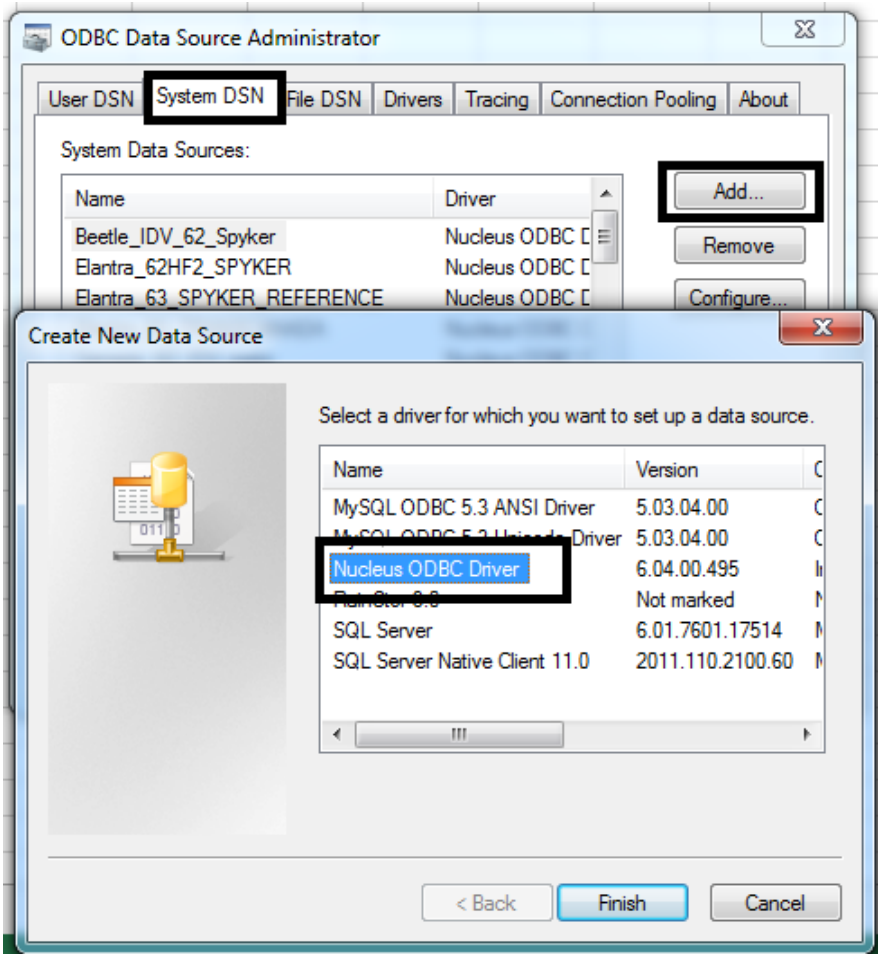
Step 2. Create the ODBC Data Source for Data Vault

Create the ODBC data source for the Data Vault archive folder that contains the required tables. After you create the data source, you can run queries on the tables that are archived into the Data Vault server.

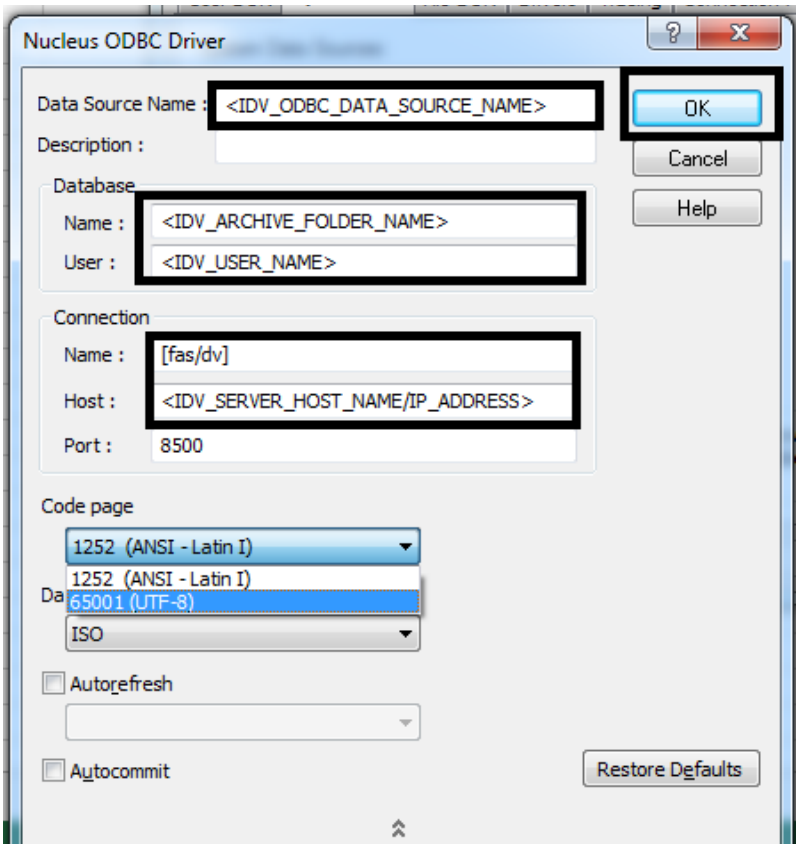
1. In the Run prompt (Windows Key+R), provide the following absolute path to launch **ODBC Data Source Administrator** program odbcad32.exe:

   For 32 bit machine: `%windir%\SysWOW64\odbcad32.exe`

   For 64 bit machine: `%windir%\System32\odbcad32.exe`
2. In the **ODBC Data Source Administrator** dialog box, click **Add** under the **System** tab to add a data source.
3. Select **Nucleus ODBC Driver**.



4. In the **Nucleus ODBC Driver** dialog box, configure the following data source properties:



You can use the following default values:

- Username. dba
- Port. 8500. The port number varies depending on the value that is configured for the Data Vault server in the `ssa.ini` file.
- Connection name. fas or dv based on the connection name that you choose when you install the Data Vault server.

To obtain the actual port value and connection name for the Data Vault server, perform the following steps:

1. Login to IDV server machine.
2. Go to [IDV_HOME] folder, where the IDV service is installed and from where it is started.
3. Open the `ssa.ini` file and check the value of attribute PORT in the server section of the file. That value is the IDV server port.
4. Once port number is obtained, open `nucleus.ini` from `[IDV_ODBC_CLIENT_INSTALL]` folder in Windows client machine to obtain ODBC connection name. The folder where IDV ODBC Administrator and Query tool is installed in Windows client machine in which ODBC DSN is being created. The `nucleus.ini` file at `[IDV_HOME]` folder can also be used to obtain ODBC connection name.
5. Confirm if there is an entry with connection `fas` or `dv`.
    a. If there is an entry with connection fas in the 'nucleus.ini' file, then ODBC connection name must be provided as 'fas' in ODBC DSN.

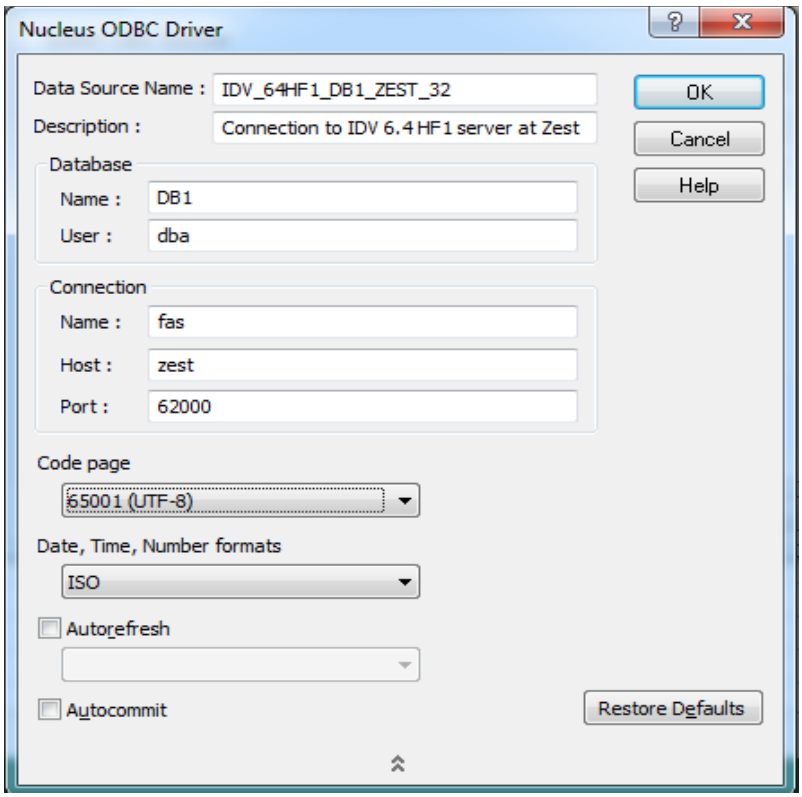
    ```
    [CONNECTION fas]
    Port=<IDV_server_port>
    ```

```
Host=<IDV_server_host>
```

b. If there is an entry with 'connection dv' as below in the 'nucleus.ini' file, then ODBC connection name must be provided as 'dv' in ODBC DSN.

```
[CONNECTION dv]
Port=<IDV_server_port>
Host=<IDV_server_host>
```

Following is the example of IDV_64HF1_DB1_ZEST_32 ODBC DSN:



Create ODBC DSN to query data in tables present in the archive folder named 'DB1' in IDV.

Archive folder is present in IDV server running at machine with hostname ‘zest' and IDV service is listening for connection requests on port 62000.

Connection is initiated with the IDV username as 'dba'. The password is not provided in DSN. When ODBC DSN is used to create connection, the password is prompted.

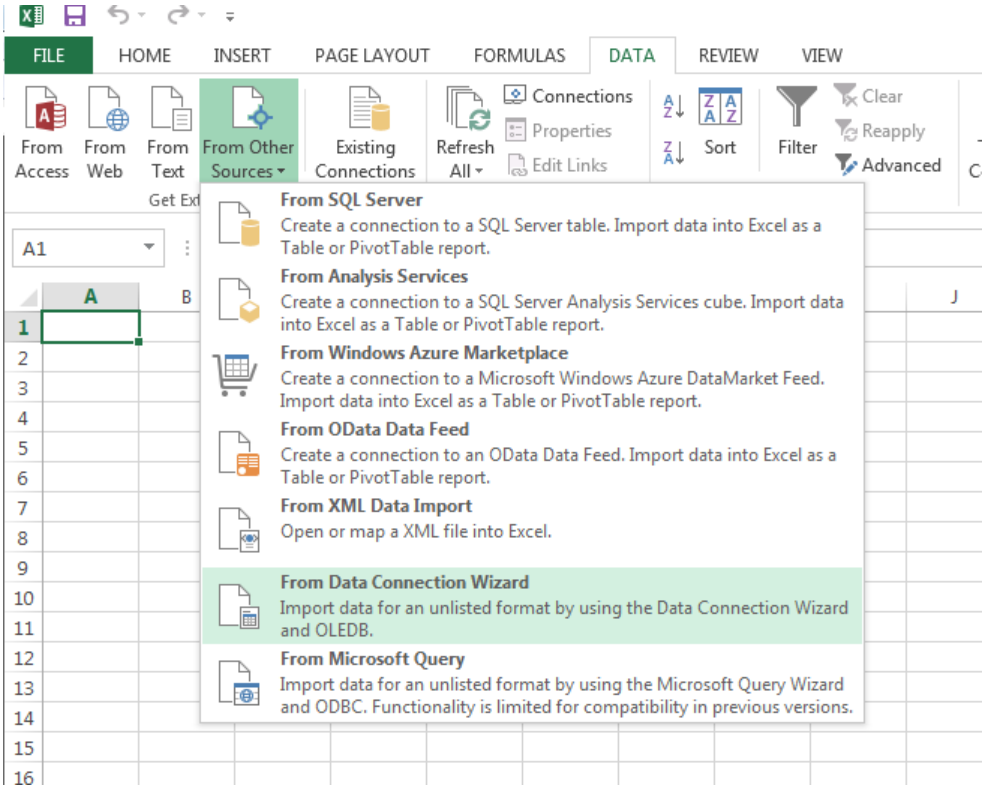
Code page for the ODBC DSN is set to UTF-8, so that UTF-8 data present in the tables at archive folder 'DB1' can be read properly.

2. After you fill all the attributes with appropriate values, click **OK** to create the DSN.
3. Repeat the process of creating DSN for each of the archive folders that has to be queried.

   Microsoft Excel is a 32-bit client. Create System DSNs with 32-bit ODBC Data Source Administrator, accessible from the `%windir%\SysWOW64\odbcad32.exe` command.

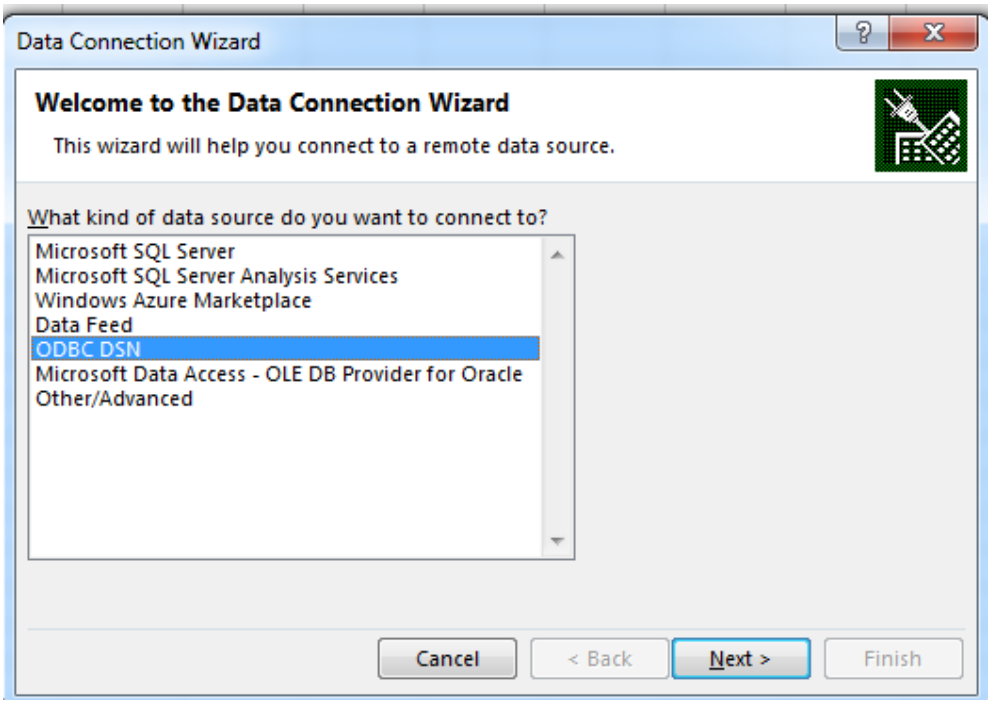
Step 3. Create Connection to Data Vault with Microsoft Excel

1. In Windows operating system, click **Start > All Programs > Microsoft Office > Microsoft Excel**.
2. Click the **DATA** menu.
3. Click **From Other Sources > From Data Connection Wizard**.



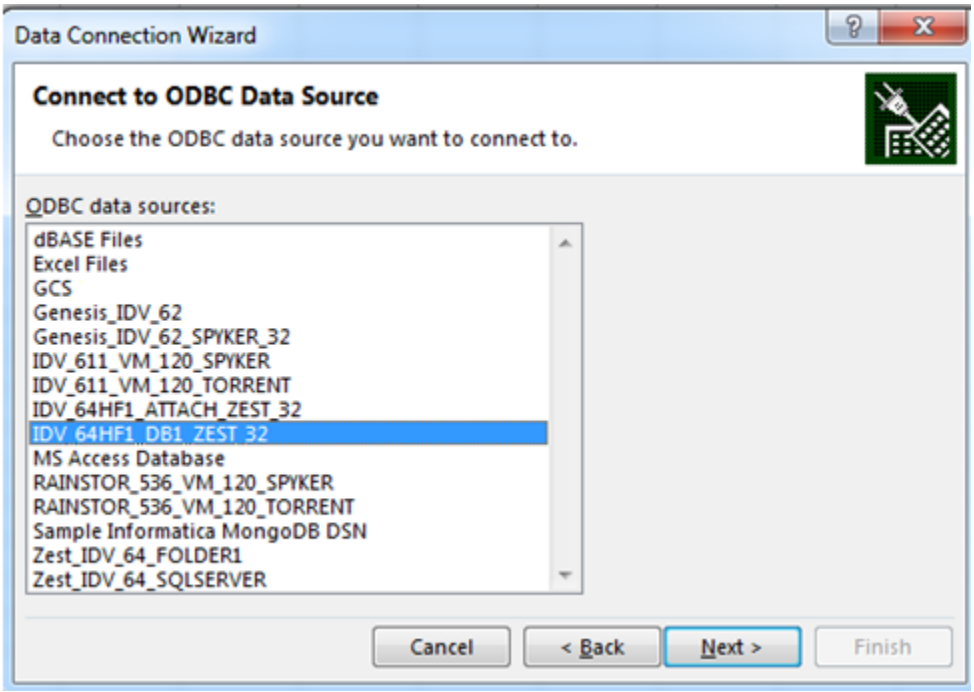
The **Data Connection Wizard** dialog box is displayed.

4. Select **ODBC DSN** and click **Next**.

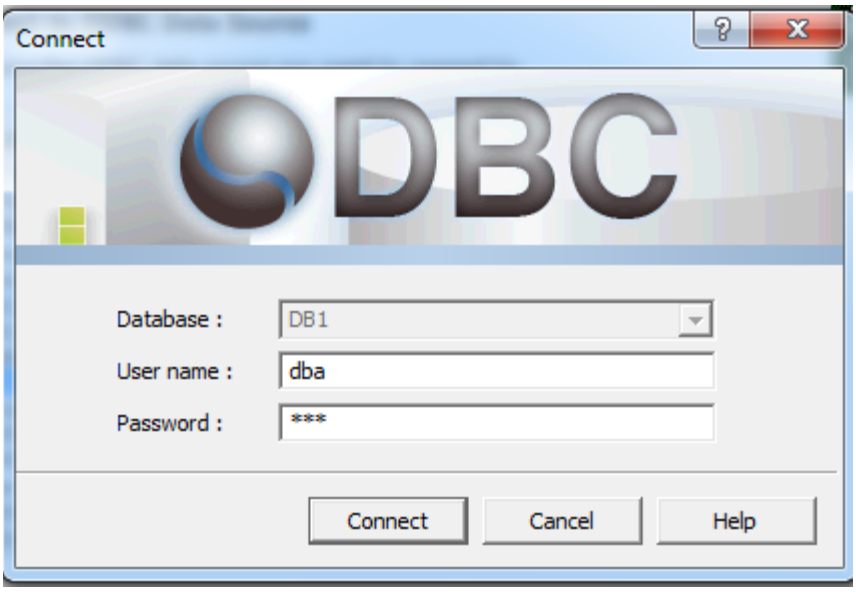


All the available ODBC System DSN's created in 32-bit ODBC Data Source Administrator client (`%windir%\SysWOW64\odbcad32.exe`) is displayed.

5. Select the ODBC DSN created to the required archive folder in IDV and click **Next**.

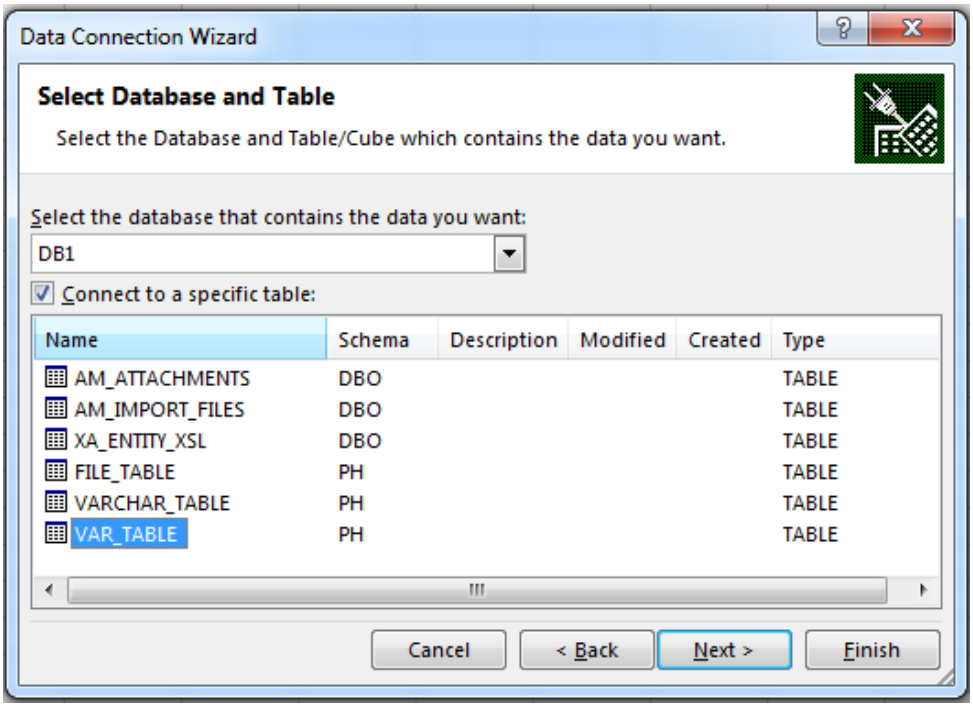


6. Enter the IDV username and password details, and click **Connect**.



If the connection window is not getting displayed, then it is due to IDV ODBC drivers (`sando.dll and sandos.dll`) missing in the '`%windir%\Syswow64`' location. Run the steps mentioned in **KB 467407** to resolve the issue and to get the connection window.

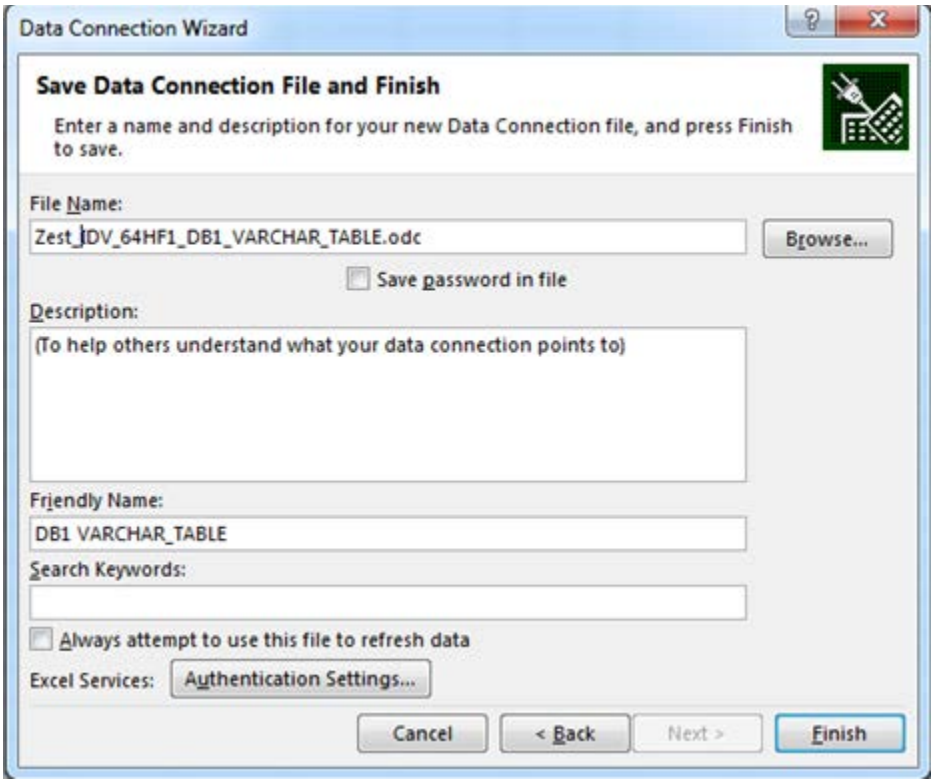
7. Once connected successfully, select the required table for which data needs to be imported into Microsoft Excel from IDV and click **Next**.



The **Save Data Connection File and Finish** page appears.

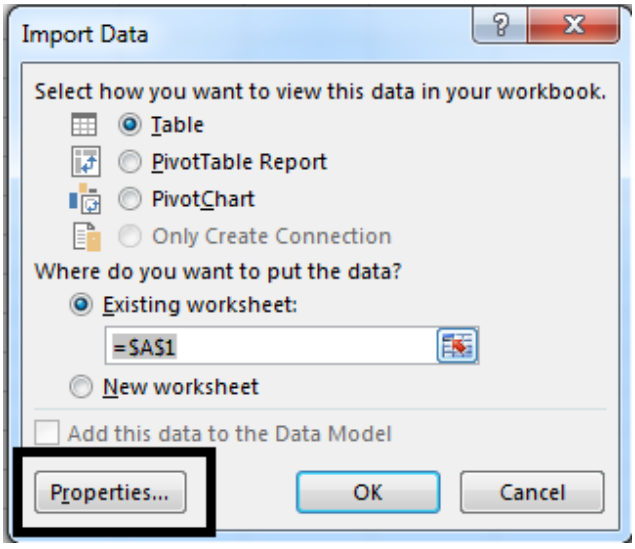
8. Provide required name to the connection and click **Finish**.

   Office Database Connection type is created.

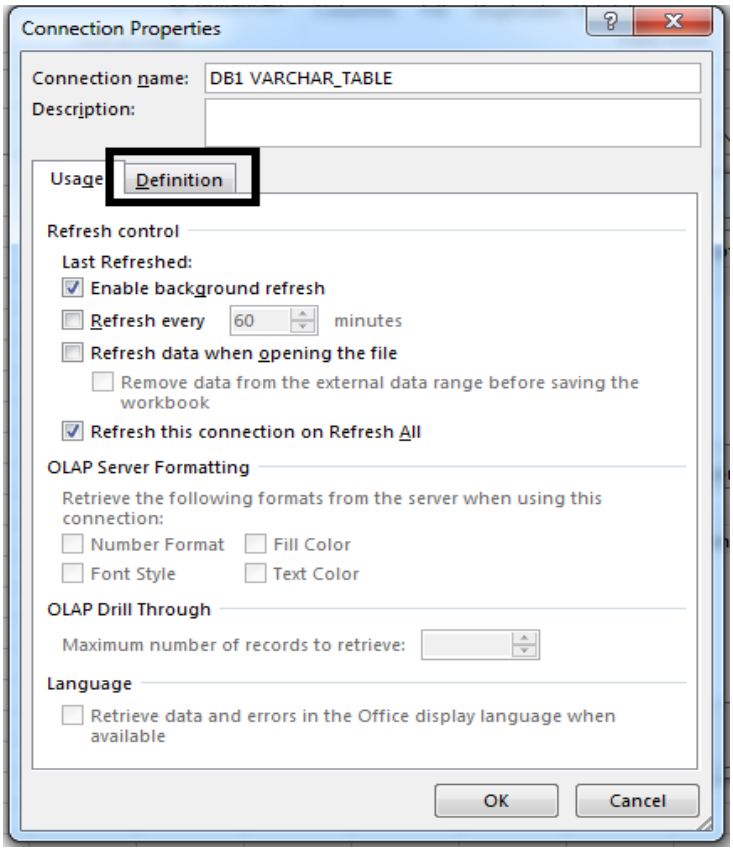


Step 4. Use Microsoft Excel to Retrieve Data

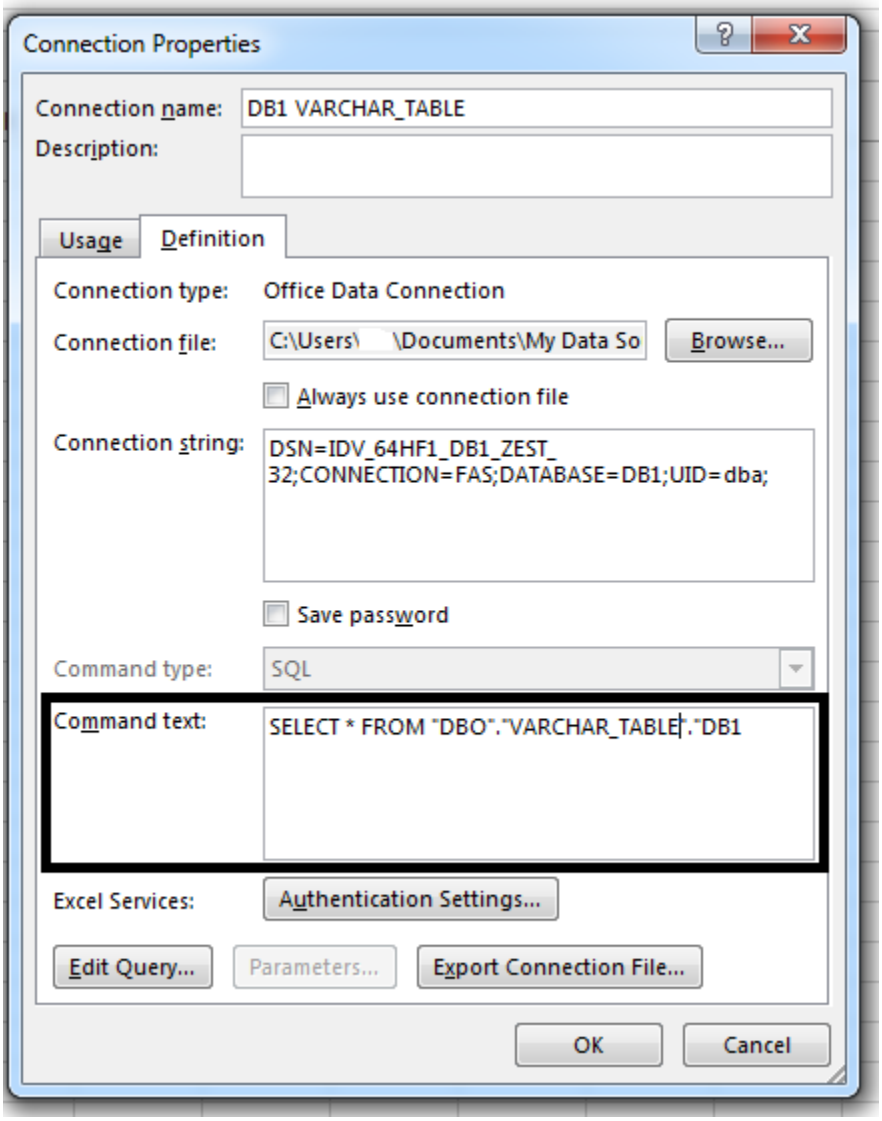
1. In the **Import Data** window, click **Properties**.



2. In the Connection Properties window, click the **Definition** tab.



3. In the **Command Text** field, enter an SQL query that you can run to extract data from the IDV archive folder into Microsoft Excel.



4. By default, Microsoft Excel uses three-way qualifier name to access database objects, such as tables. In IDV, you must use two-way qualifier name to access the database objects. If you use three-way qualifier names to access objects in an SQL query, then query execution fails with parser error.

   For example, to access a table named 'TICKET' in a schema named 'DBO' of database named 'LOYALTY', you can provide the following three-way and two-way qualifier names:

   - Three-way qualifier name. LOYALTY.DBO.TICKET
   - Two-way qualifier name. DBO.TICKET

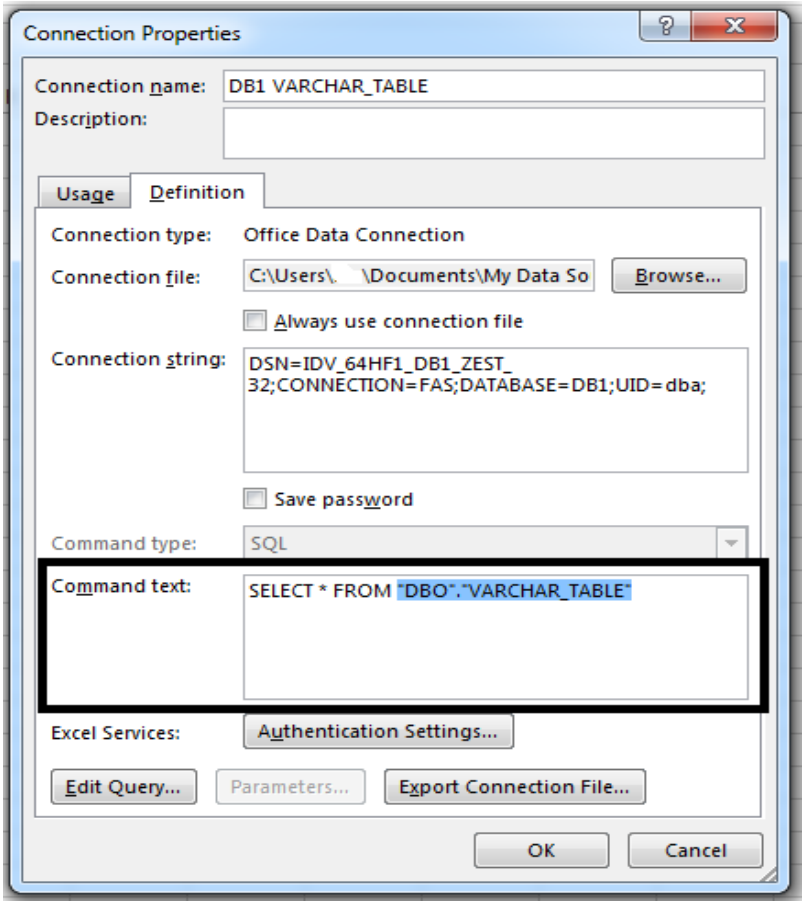
   Sometimes, Microsoft Excel can add a database name after a table name.

   In the Command Text field, change the three-way table name qualifier in the SQL query to a two-way table name qualifier in the following format: SCHEMA_NAME.TABLE_NAME

   After you add the two-way qualifier name, add the required filter conditions to SQL query. Filter conditions can involve multiple columns in the same table.

   For information on filtering query results, see Filtering Query Result Set in Data Vault

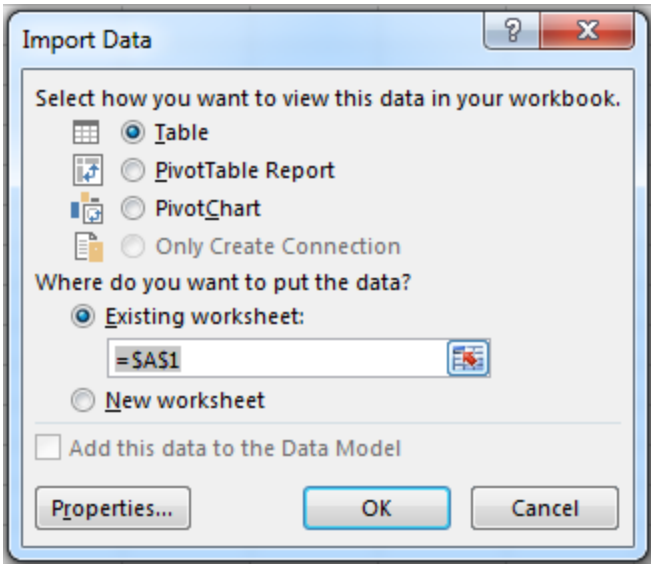
5. To save the changes, click **OK**.



The IDV ODBC 'Database Query' type connection is created in Microsoft Excel Worksheet with the name provided in the 'Connection Name' field of the 'Connection Properties' window. Created connection contains details such as IDV ODBC DSN information and SQL Query used for fetching data from IDV.

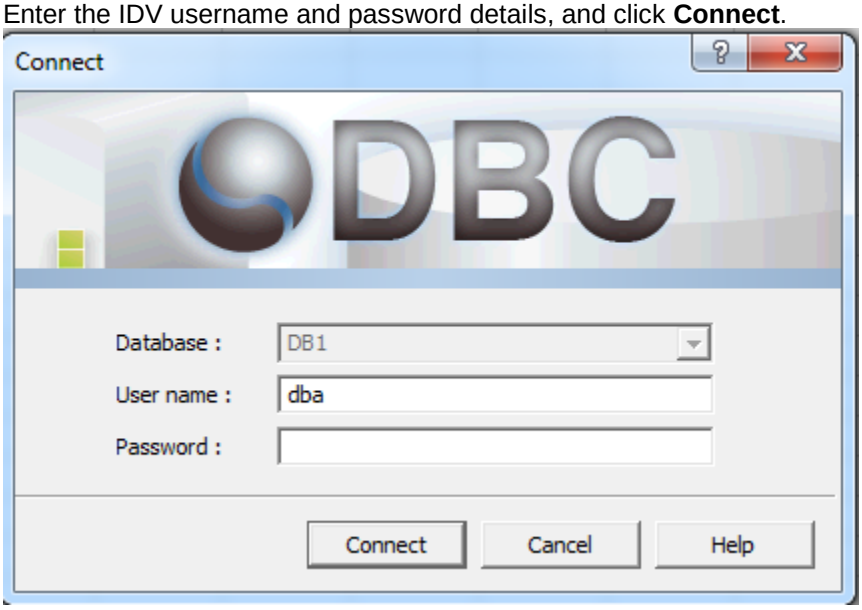
After the table name qualifier is updated to 'SCHEMA_NAME.TABLE_NAME' and you click **OK**, the **Import Data** dialog box is displayed again.

6. To create the connection to IDV, click **OK**.



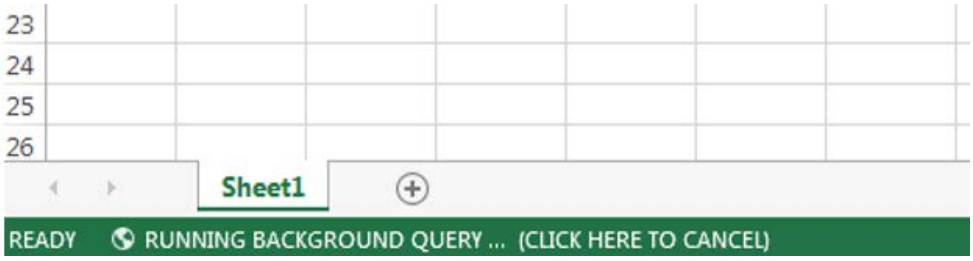
The IDV connection dialog box is displayed.

7. Enter the IDV username and password details, and click **Connect**.

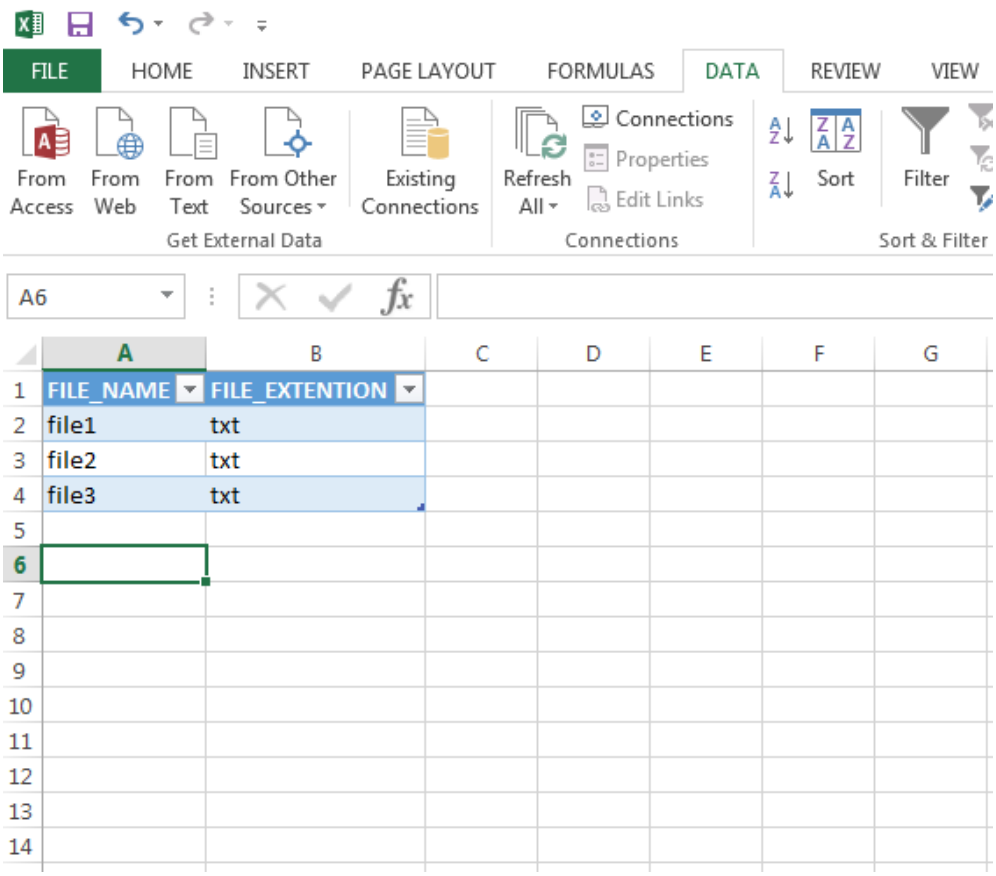


Microsoft Excel initiates the connection to Data Vault and query execution starts in the Data Vault.

When query execution is in progress, status is displayed at the bottom of Microsoft Excel worksheet. The SELECT query execution takes time depending on the amount of the result set to be returned.



8. When query execution completes, results are displayed in a tabular form in the Microsoft Excel worksheet as shown below:


FILE
HOME
INSERT
PAGE LAYOUT
FORMULAS
DATA
REVIEW
VIEW
From Access
From Web
From Text
From Other Sources
Existing Connections
Refresh All
Connections
Properties
Edit Links
Sort
Filter
Get External Data
Connections
Sort & Filter
A6
A
B
C
D
E
F
G
1 FILE_NAME FILE_EXTENTION
2 file1 txt
3 file2 txt
4 file3 txt
5
6
7
8
9
10
11
12
13
14

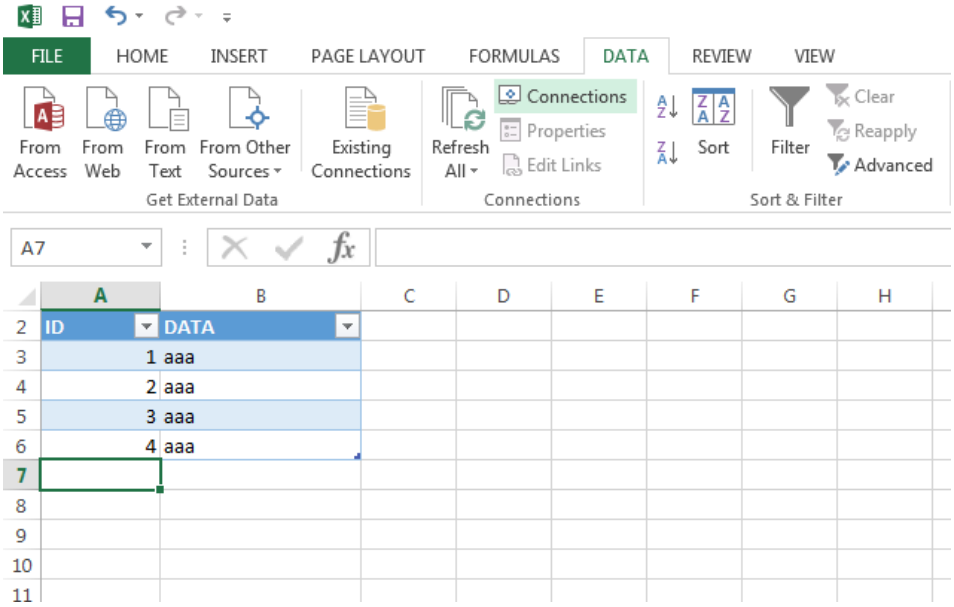
Additional Functionalities

You can perform multiple operations when you use Microsoft Excel to extract data from Data Vault. You can filter query results to restrict the data from Data Vault. You can reuse the IDV ODBC SQL queries across Microsoft Excel.

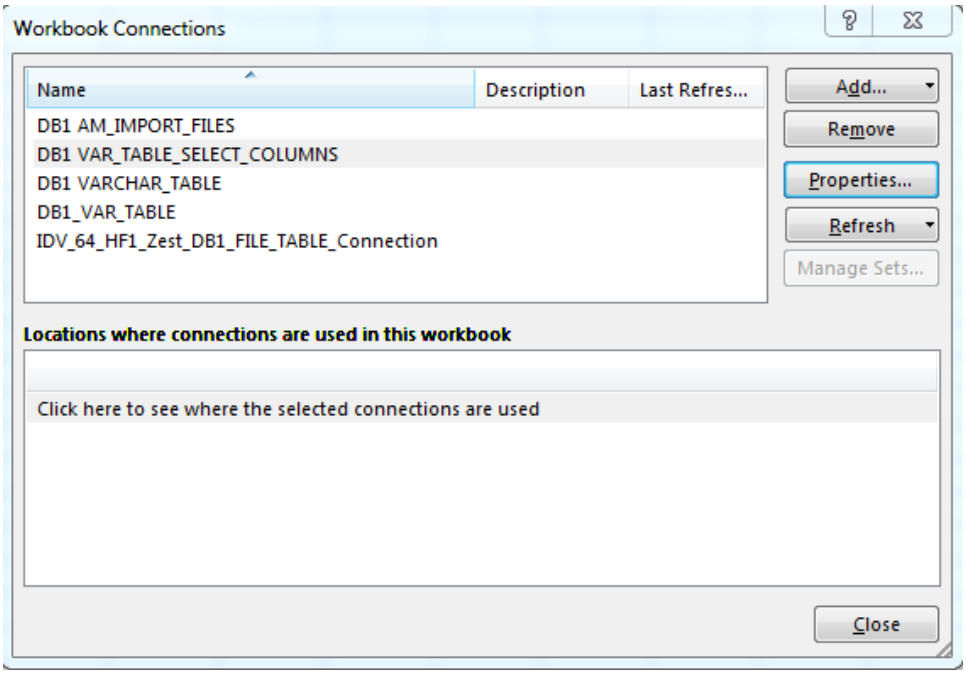
Filtering Query Result Set in Data Vault

You can restrict result data set for the table at IDV while you retrieve data into Microsoft Excel. For result set data filtering at IDV, provide filtering conditions in the 'Definition' section of the 'Connection Properties' window of Microsoft Excel, where IDV table name qualifier is updated to a 'two-way qualifier'.

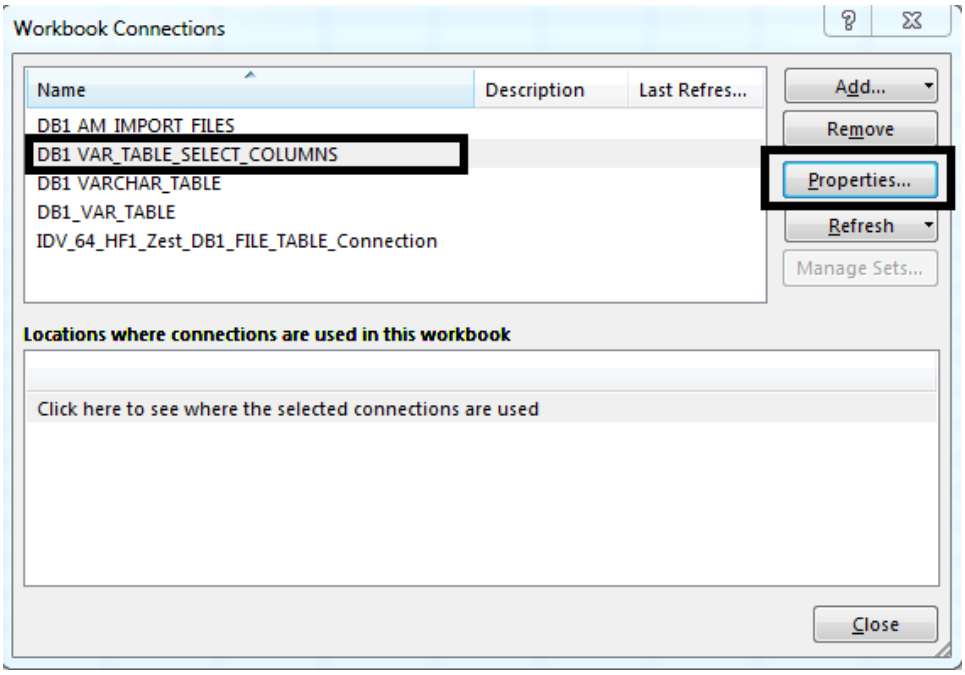
1. To launch Microsoft Excel, click **Start > All Programs > Microsoft Office > Microsoft Excel**.
2. Open Excel Workbook, where the IDV ODBC SQL query is created and data is fetched from IDV.
3. Click the **DATA** menu.
4. Click **Connections**.



The **Workbook Connections** window opens. The window contains all types of connections that are used in the active workbook, including the required IDV ODBC connections containing SQL queries used in fetching data into the Microsoft Excel worksheet.

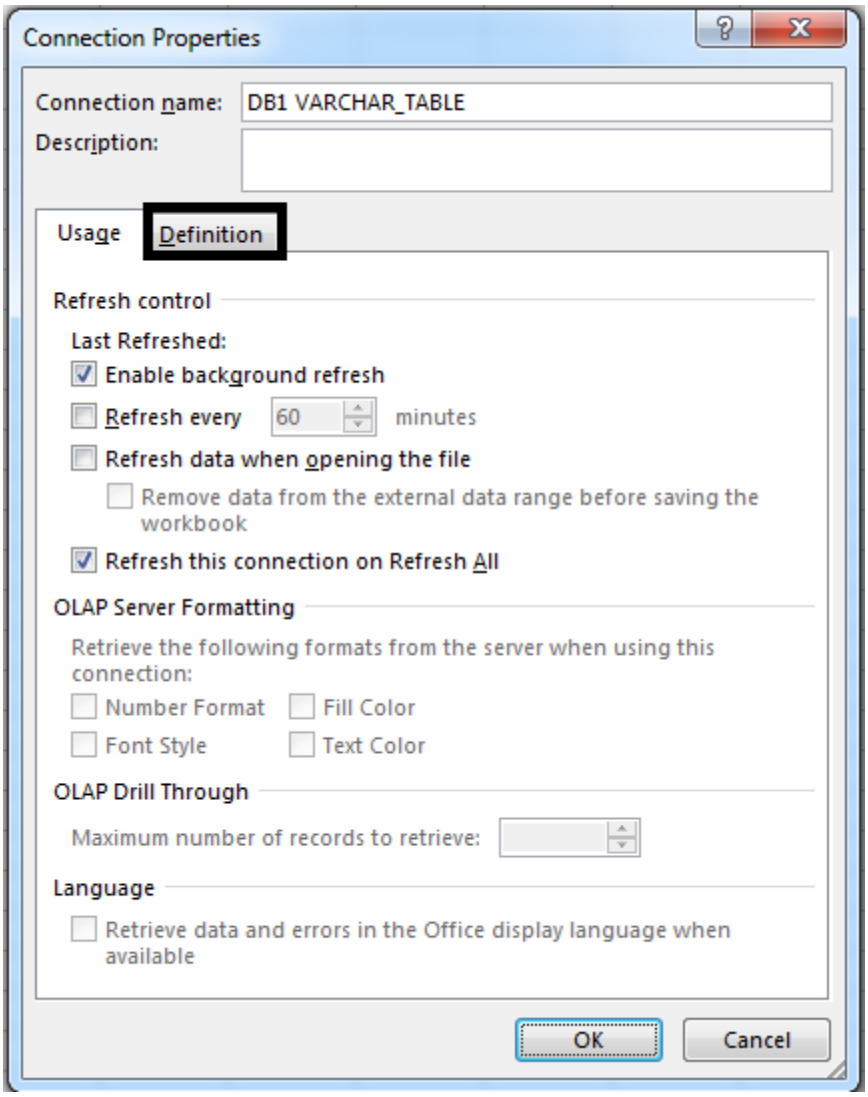


5. Select the required **IDV ODBC Connection** and click **Properties**.

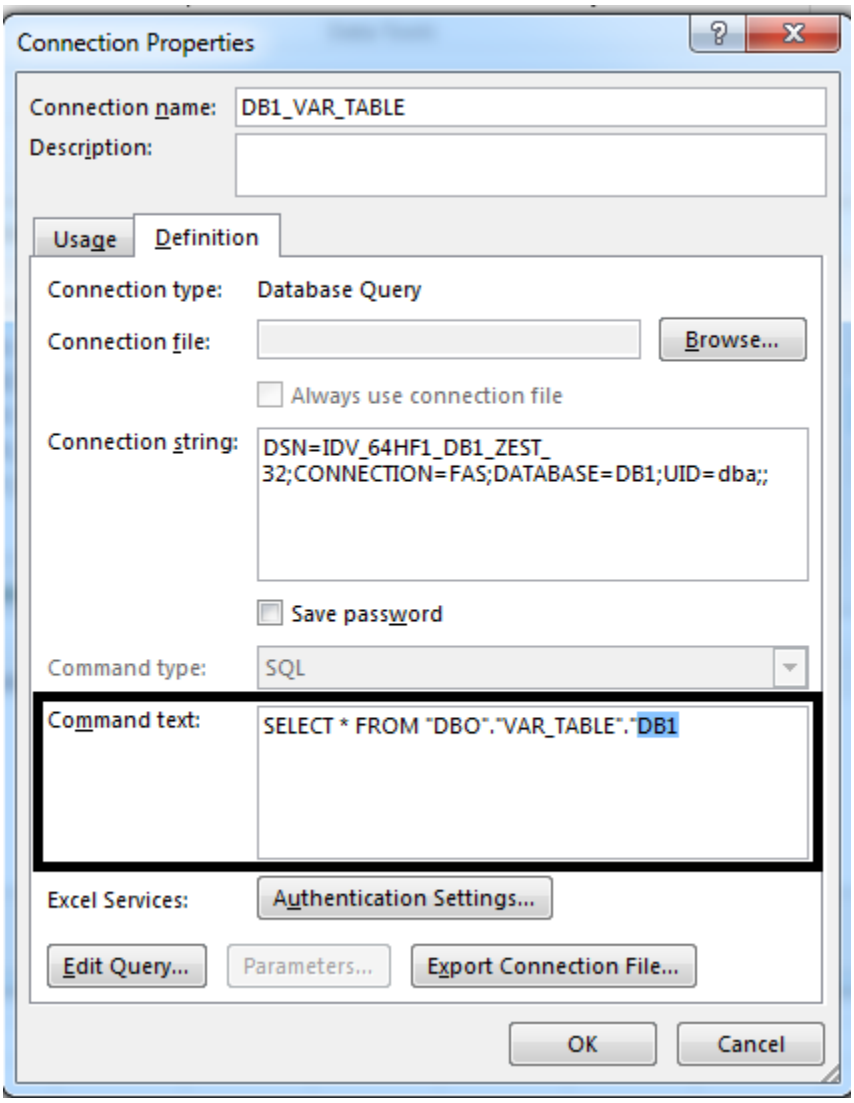


The **Connection Properties** window with properties such as 'ODBC DSN' and 'SQL Query' for the selected 'IDV ODBC Connection' are displayed.

6. Click the **Definition** tab.



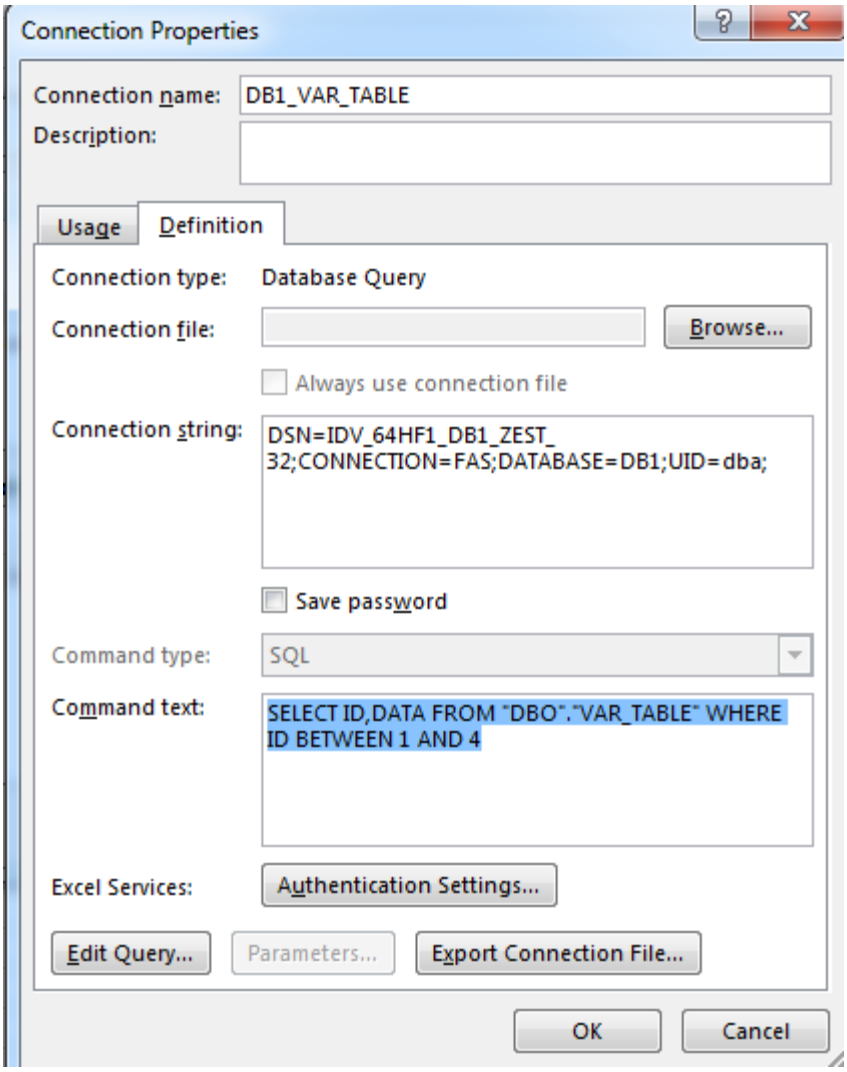
7. In the 'Command Text' field, verify the SQL query that is executed in the IDV archive folder to fetch data into Microsoft Excel.



By default, 'SELECT *' query is executed in IDV by Microsoft Excel on the selected table and table name qualifier used in query is a three-way qualifier.

After you update the table name qualifier to a two-way qualifier, apply the required filtering conditions. Filtering conditions can involve multiple columns in the same table.

An example of the SELECT query that is used to restrict the result set for 'VARCHAR_TABLE' at IDV using filtering condition on 'ID' column is displayed below:



8. To save the changes to the connection, click **OK**.

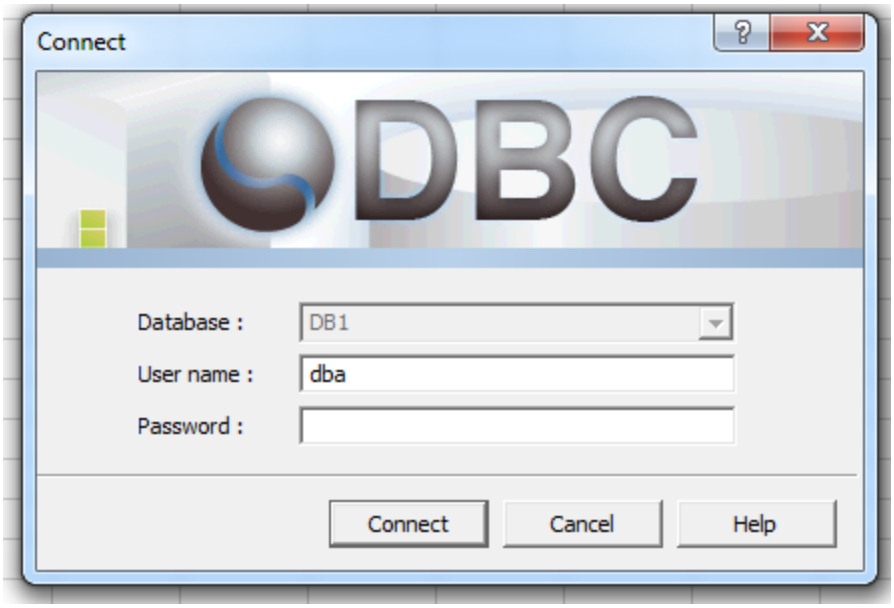
   IDV ODBC Connection is saved in Microsoft Excel worksheet with name provided in 'Connection Name' text field of the 'Connection Properties' window. Created connection contains the IDV ODBC DSN details and SQL query used for fetching data.

9. In **Import Data** dialog box, click **OK**.

   The IDV connection window is displayed.

10. Enter the IDV username and password details, and click Connect.

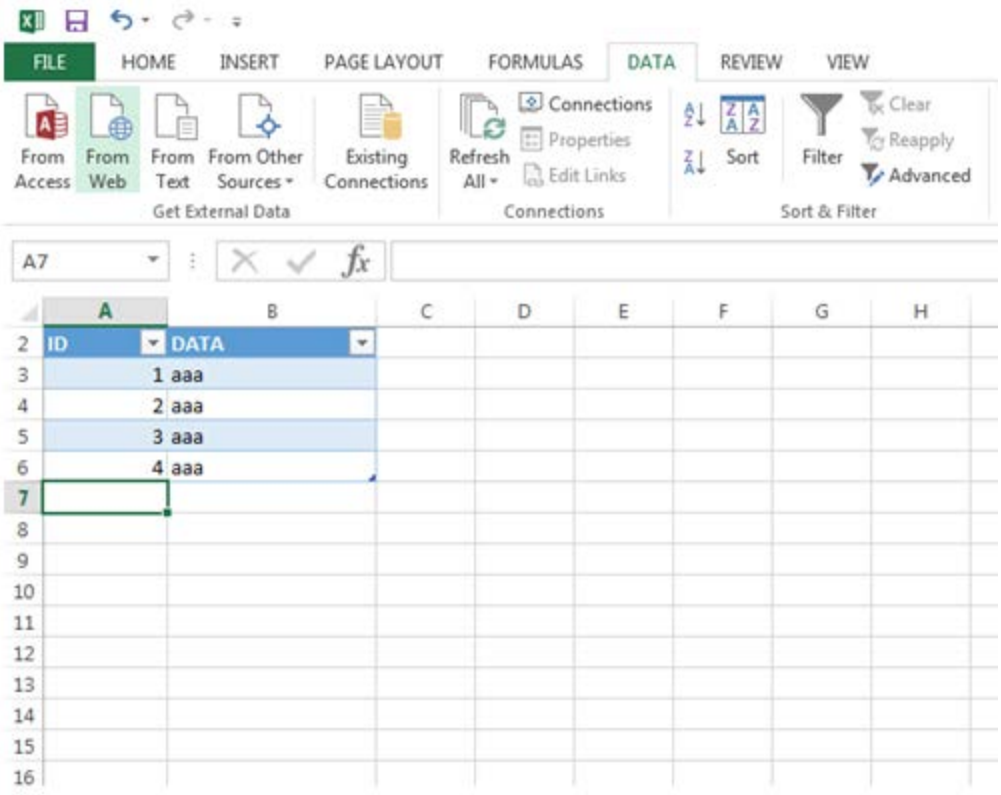
    Microsoft Excel initiates the connection to Data Vault and query execution starts in the Data Vault.



11. When query execution is in progress, status is displayed at the bottom of Microsoft Excel worksheet. The SELECT query execution takes time depending on the amount of the result set to be returned.



12. When query execution completes, results are displayed in a tabular form in the Microsoft Excel worksheet as shown below:



Reusing IDV ODBC SQL Queries across Microsoft Excel Workbooks

When data of a required table is retrieved into Microsoft Excel using SQL queries, a default connection of type Database Query is created automatically by Microsoft Excel. The Database Query type connection cannot be used outside the Microsoft Excel Workbook, where it is created.

If you use SQL queries to fetch data from IDV into Microsoft Excel across different workbooks, it is required to export the 'Database Query' connection type as Office Database Connection. Once exported successfully, the 'Office Database Connection' type file can be used in different workbooks to retrieve the same data.

The Office Database Connection file works without any issues when you use it in Microsoft Excel present in same Windows client machine from where the file is exported. You can use it directly in different workbooks in that machine without any additional configuration.

You can use the exported Office Database Connection in Microsoft Excel at different Windows client machine without any additional configuration. As a prerequisite, you must install IDV ODBC Driver and create the IDV ODBC DSN in target Windows client machine with same name as in the source Windows client machine.

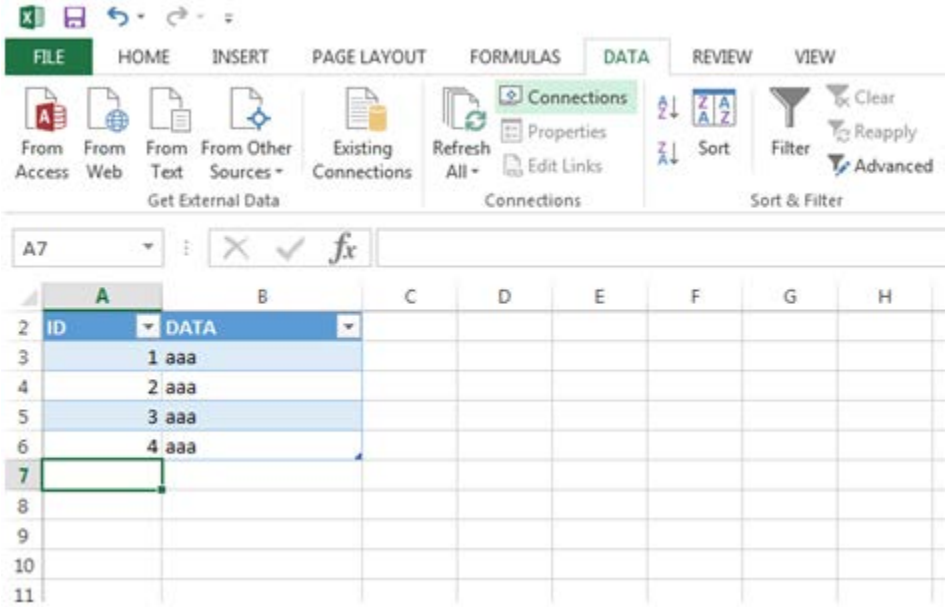
To reuse the IDV ODBC SQL queries across Microsoft Excel workbooks, perform the following steps:

- Export IDV ODBC SQL queries in Microsoft Excel workbook as Office Database Connection.

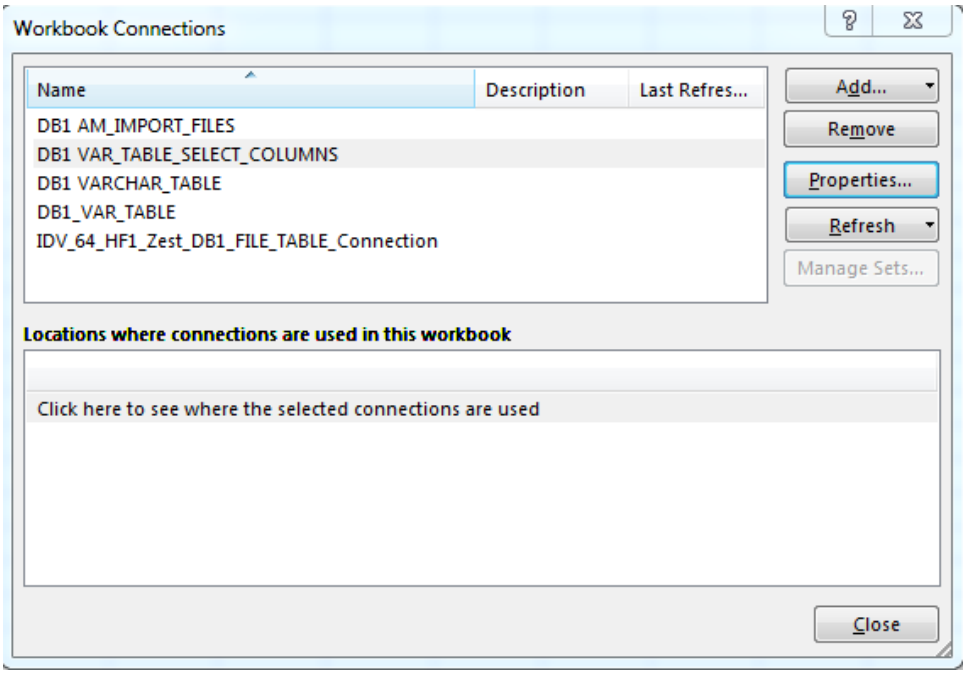
- Import IDV ODBC SQL queries in Microsoft Excel workbook using Office Database Connection.

Export IDV ODBC SQL Queries in Microsoft Excel Workbook

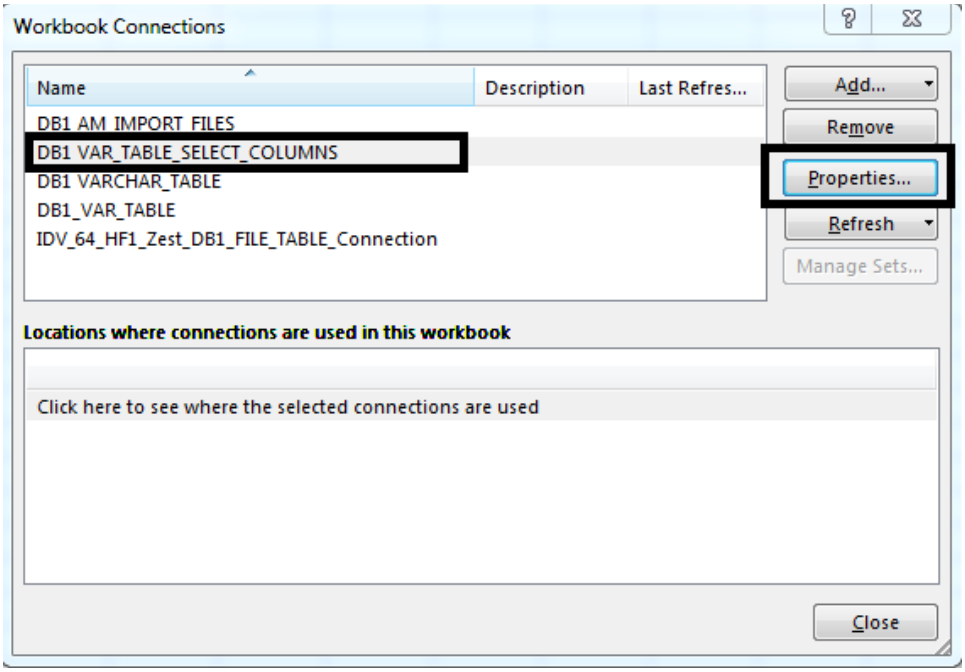
1. To launch Microsoft Excel, click **Start > All Programs > Microsoft Office > Microsoft Excel**.
2. Open Excel Workbook, where the IDV ODBC SQL query is created and data is fetched from IDV.
3. Click the **DATA** menu.
4. Click **Connections**.



The **Workbook Connections** window opens. The window contains all types of connections that are used in the active workbook, including the required IDV ODBC connections containing SQL queries used in fetching data into the Microsoft Excel worksheet.

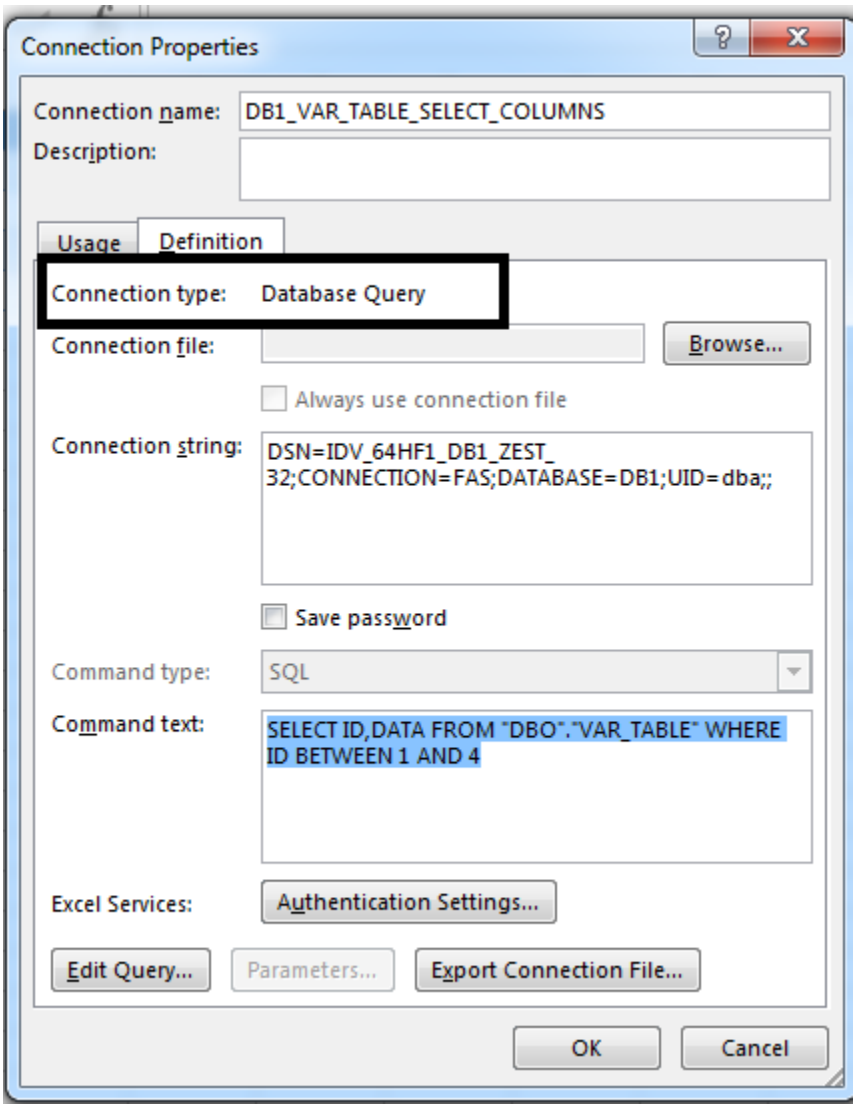


5. Select the required **IDV ODBC Connection** and click **Properties**.

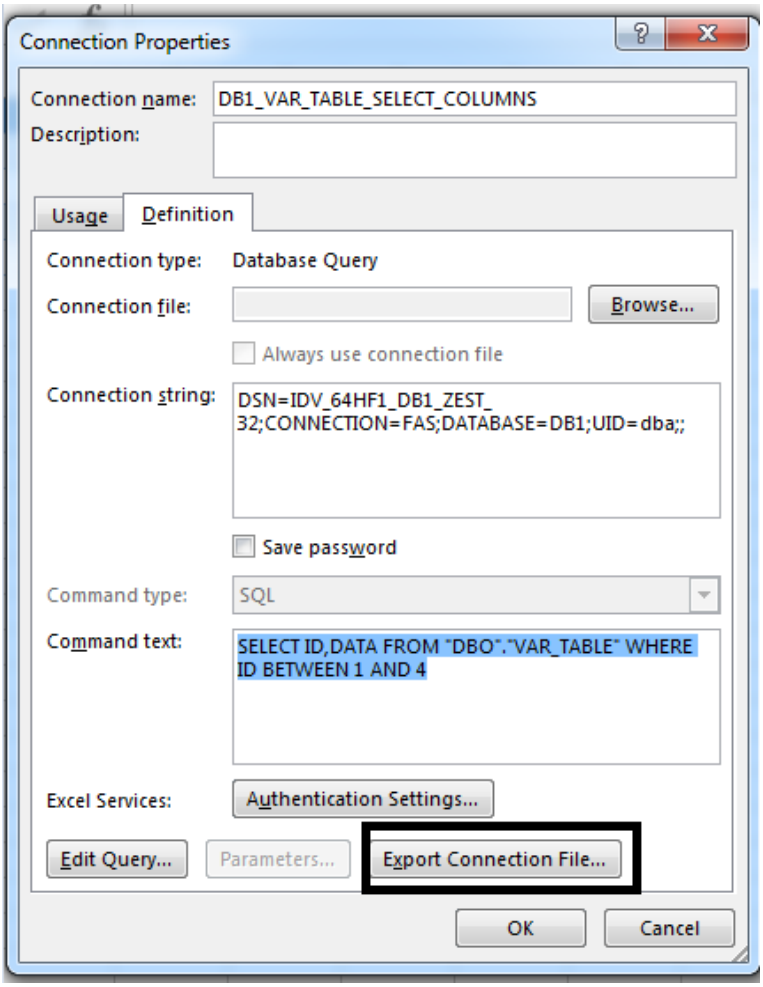


The **Connection Properties** window with properties such as 'ODBC DSN' and 'SQL Query' for the selected 'IDV ODBC Connection' are displayed.

In the 'Connection Properties' window, the 'Connection type' is 'Database Query' for IDV ODBC connections in the worksheet.



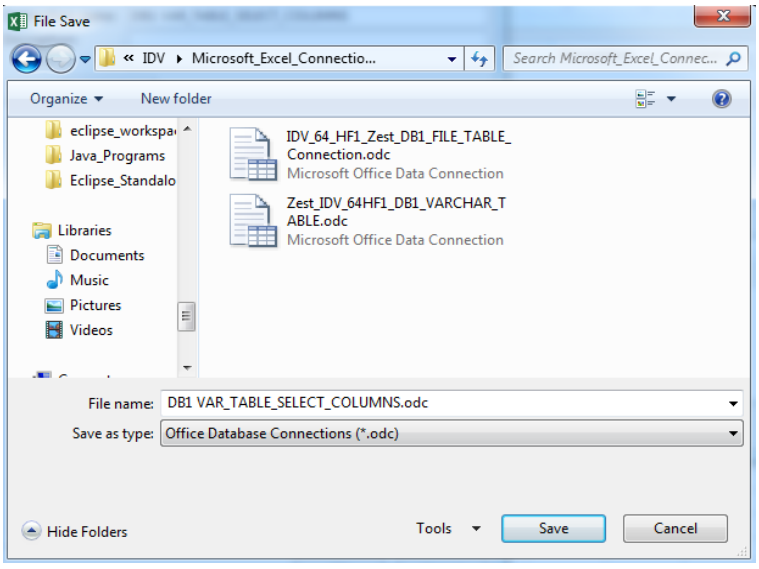
6. Click the **Definition** tab.
7. Click **Export Connection File**.



The **File Save** dialog box is displayed.

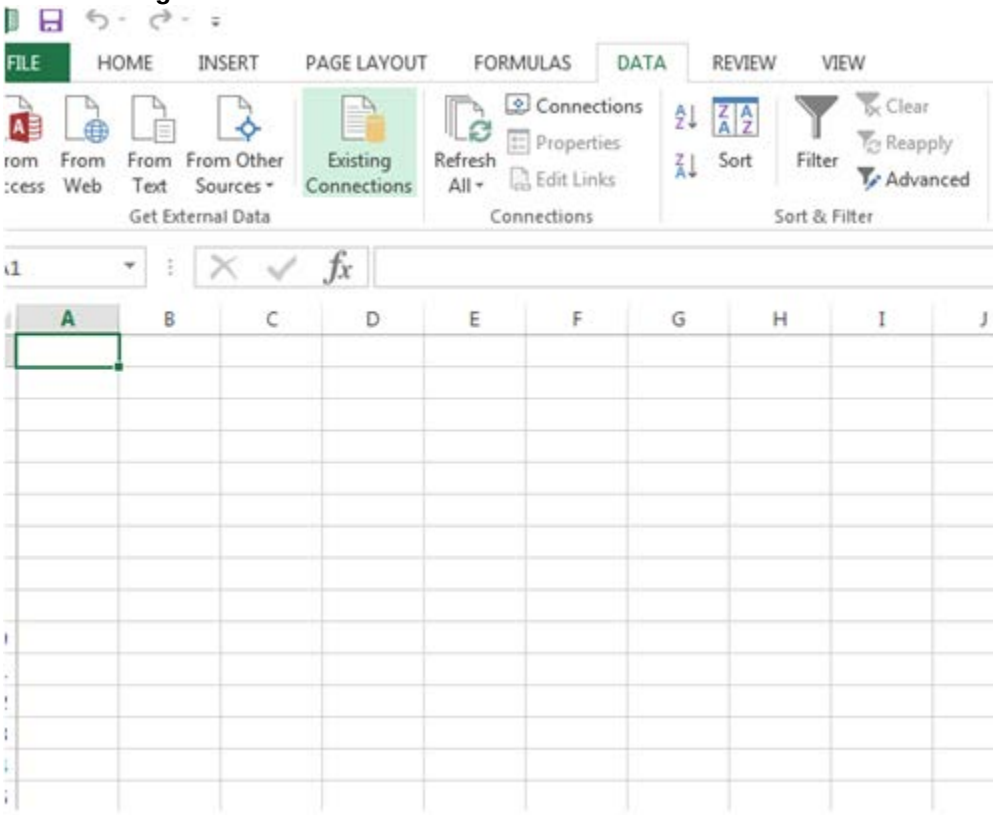
8. Save the connection at desired location with a required name to complete export of 'IDV ODBC SQL Query' connection from the workbook.

   The connection is saved as 'Office Database Connection' type.

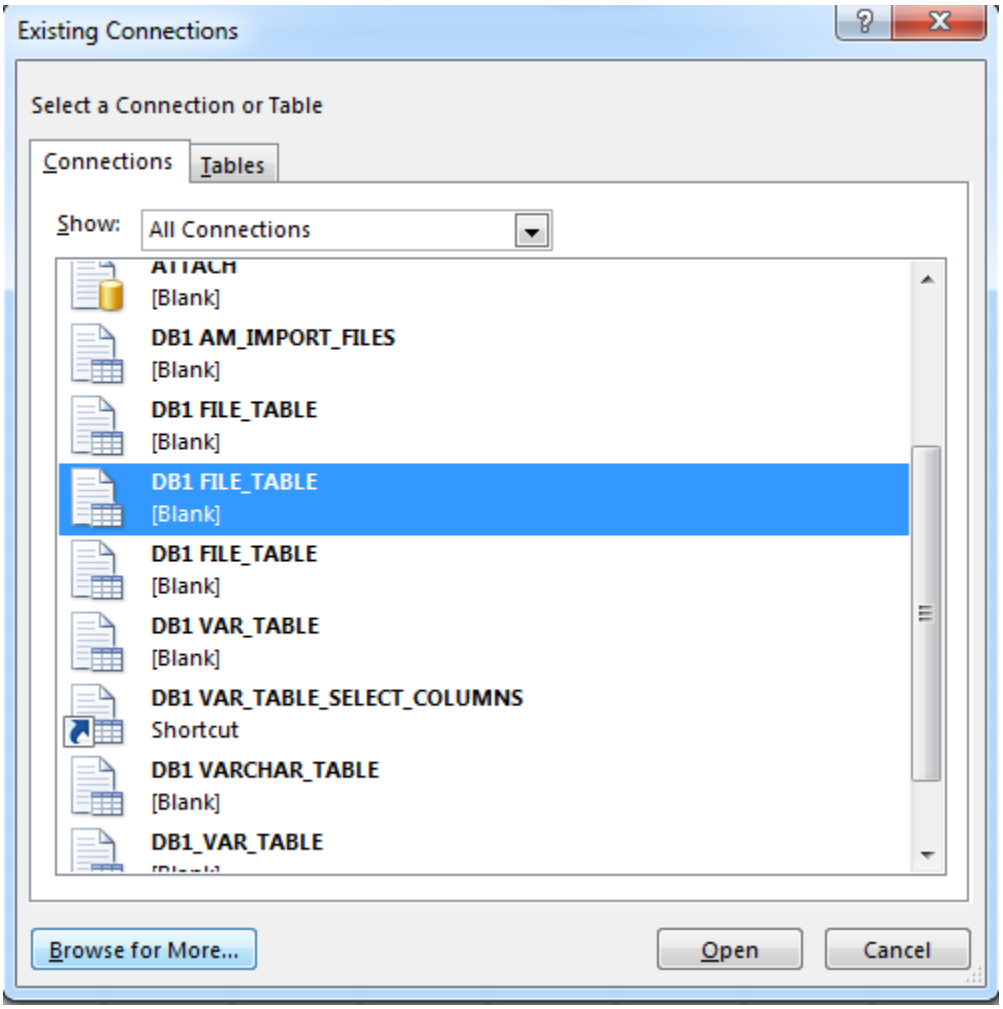


Import IDV ODBC SQL Queries in Microsoft Excel Workbook

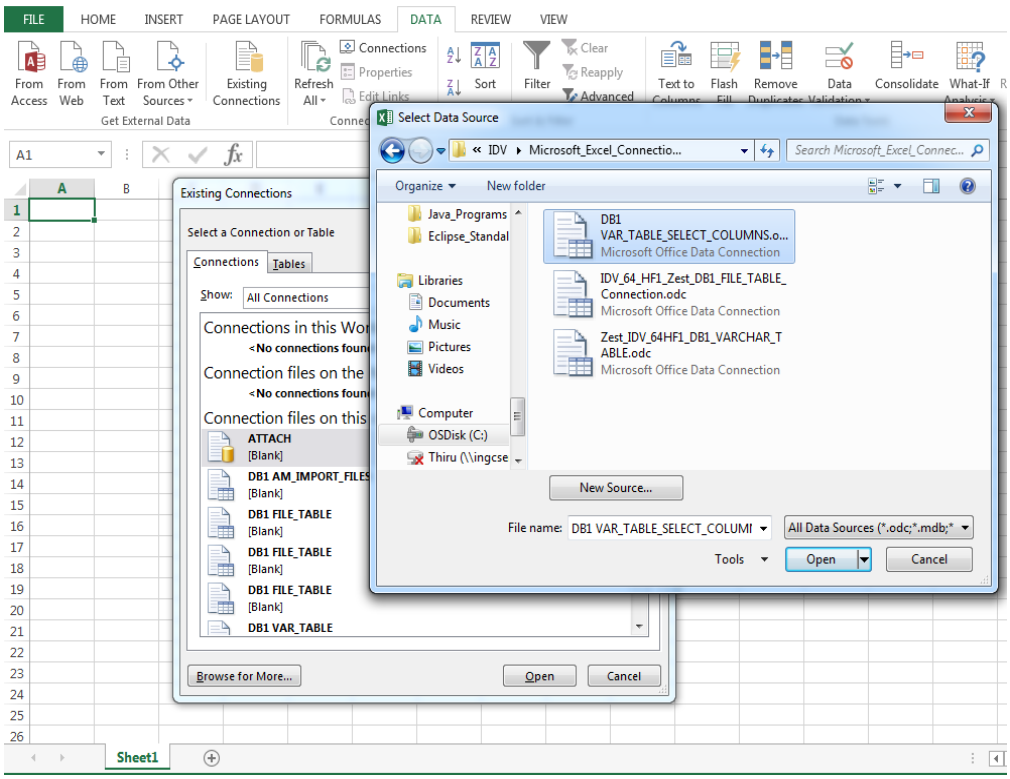
1. To launch Microsoft Excel, click **Start > All Programs > Microsoft Office > Microsoft Excel**.
2. Open a target workbook where the exported 'Office Database Connection' created for IDV ODBC SQL Queries has to be reused.
3. Add a new blank worksheet if the Excel Workbook already contains other data to clearly segregate data.
4. Click the **DATA** menu.
5. Click **Existing Connections**.



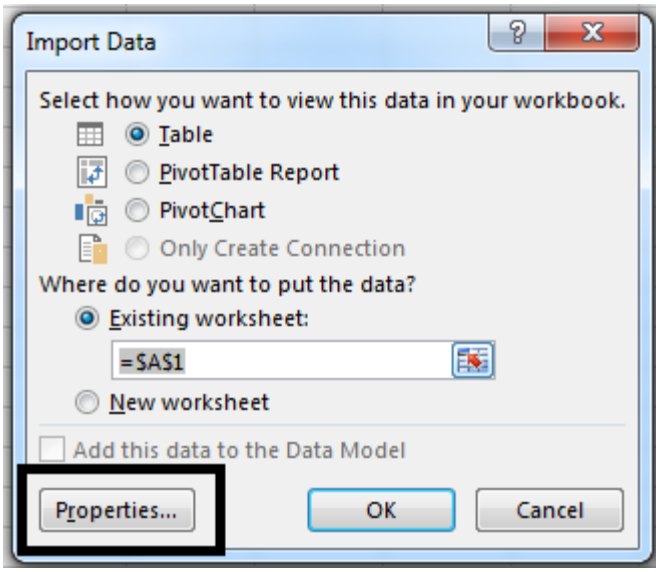
The Existing Connections window is displayed with different data source connections available in the `%userprofile%\Documents\My Data Sources` folder.



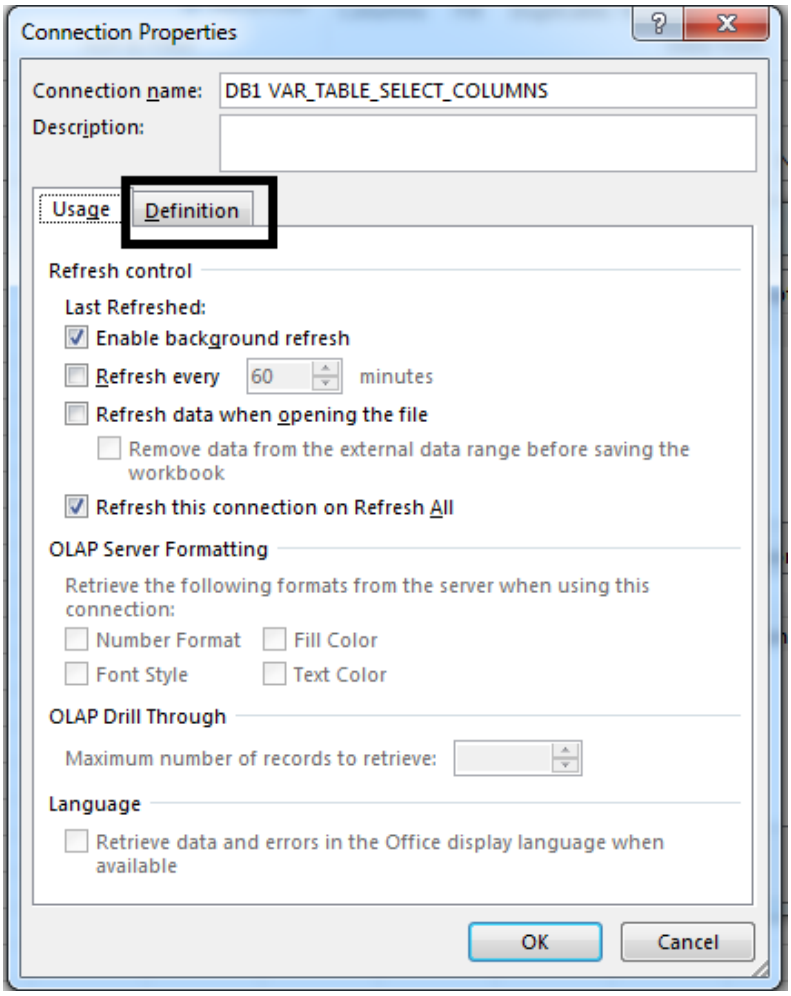
1. If the Office Database Connection type containing IDV archive folder connection details and SQL query is saved in the default location ('%userprofile%\Documents\My Data Sources'), then it is displayed. Select the connection file from the available list of connections.
2. If the Office Database Connection type with IDV ODBC SQL query details is exported into a different location than the default one ('%userprofile%\Documents\My Data Sources'), then click the **Browse for More** button to navigate to the location, where you can find the connection export file. Select the connection file.



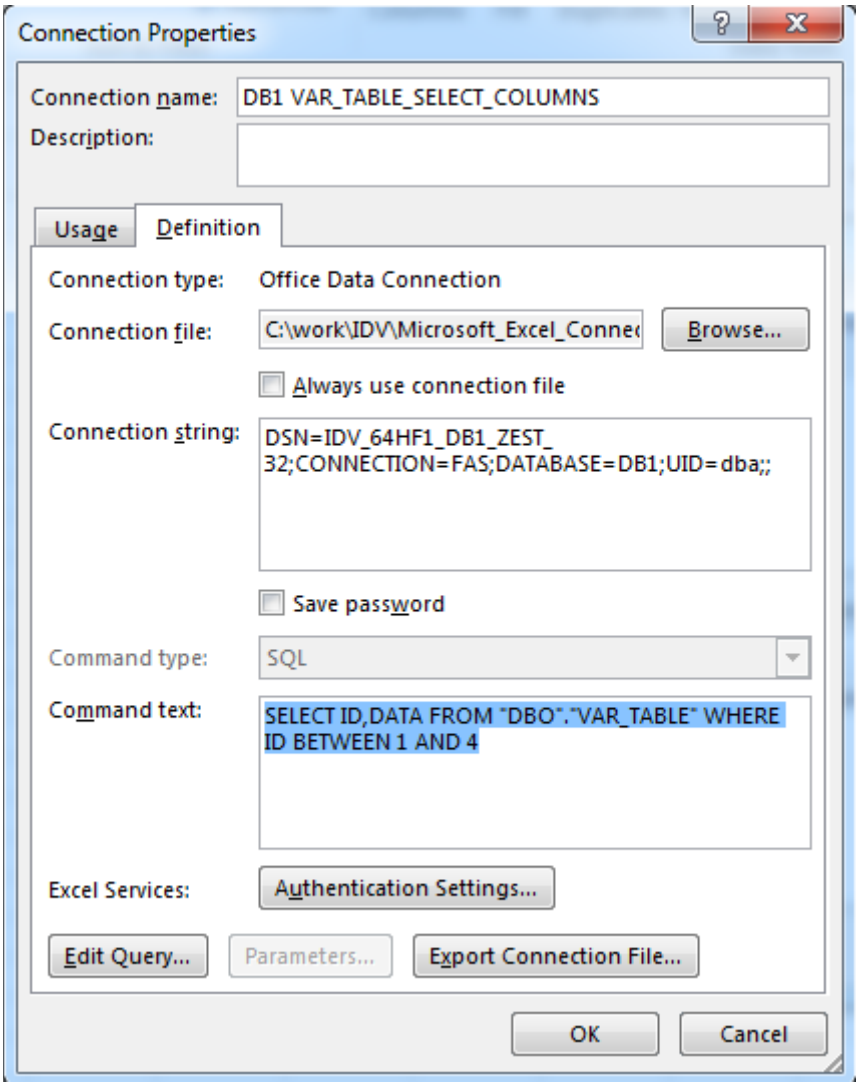
6. To use the connection in the new Excel Workbook, click **Open**.
   The **Import Data** window is displayed.
7. Click **Properties**.



   The **Connection Properties** window is displayed.
8. Click the **Definition** tab.

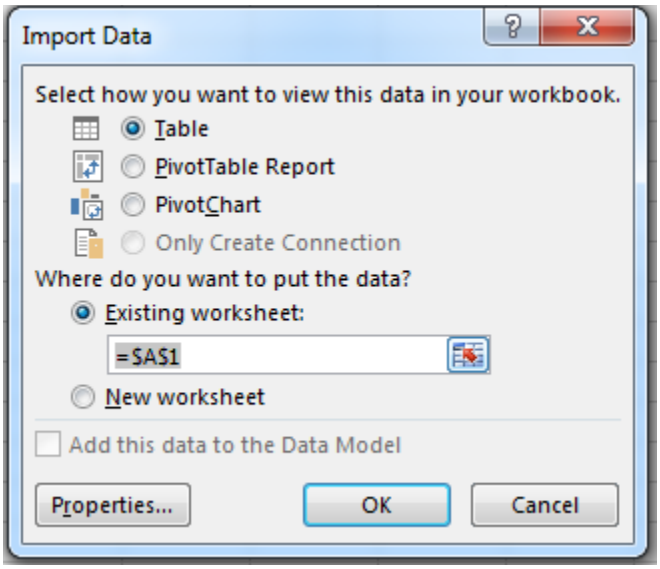


9. Verify that the available SQL query is the same as the one used earlier.
10. Click **OK**.



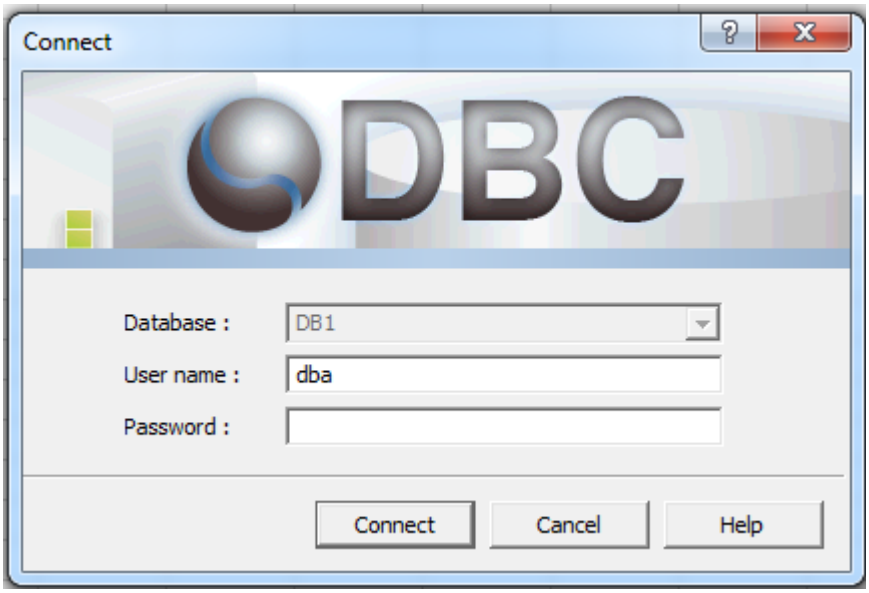
The **Import Data** dialog box is displayed.

11. To create the connection to IDV, click **OK**.



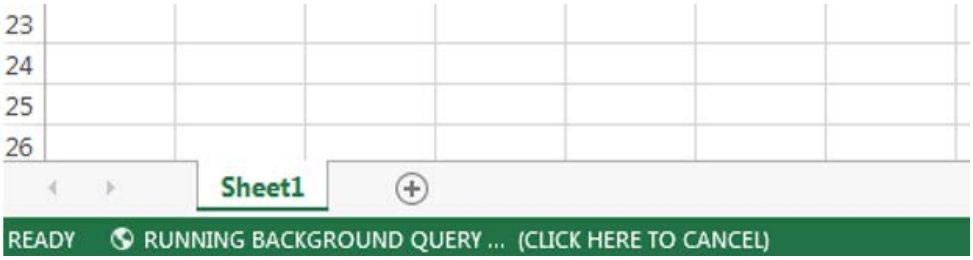
The IDV connection dialog box is displayed.

12. Enter the IDV username and password details, and click **Connect**.

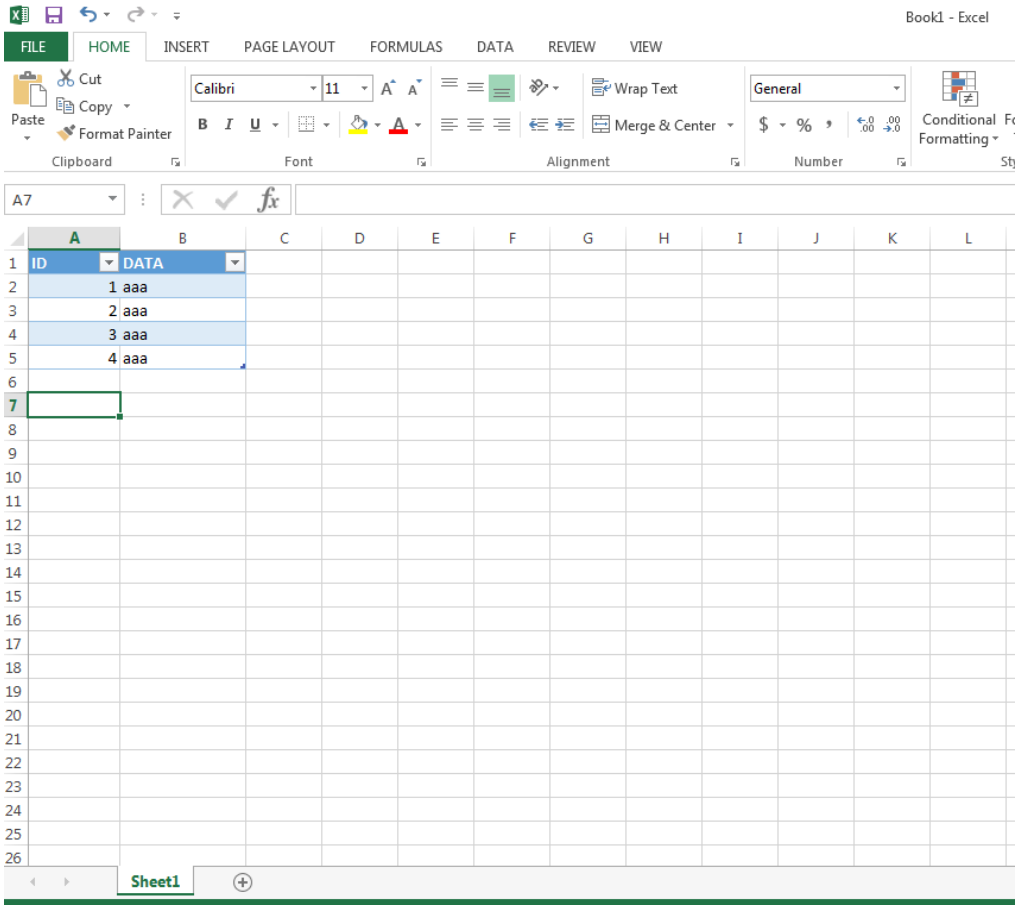


Microsoft Excel initiates the connection to Data Vault and query execution starts in the Data Vault.

When query execution is in progress, status is displayed at the bottom of Microsoft Excel worksheet. The SELECT query execution takes time depending on the amount of the result set to be returned.



13. When query execution completes, results are displayed in a tabular form in the Microsoft Excel worksheet as shown below:



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